

A Visual Analytic Environment to Co-locate Peoples' Tweets with City Factual Data



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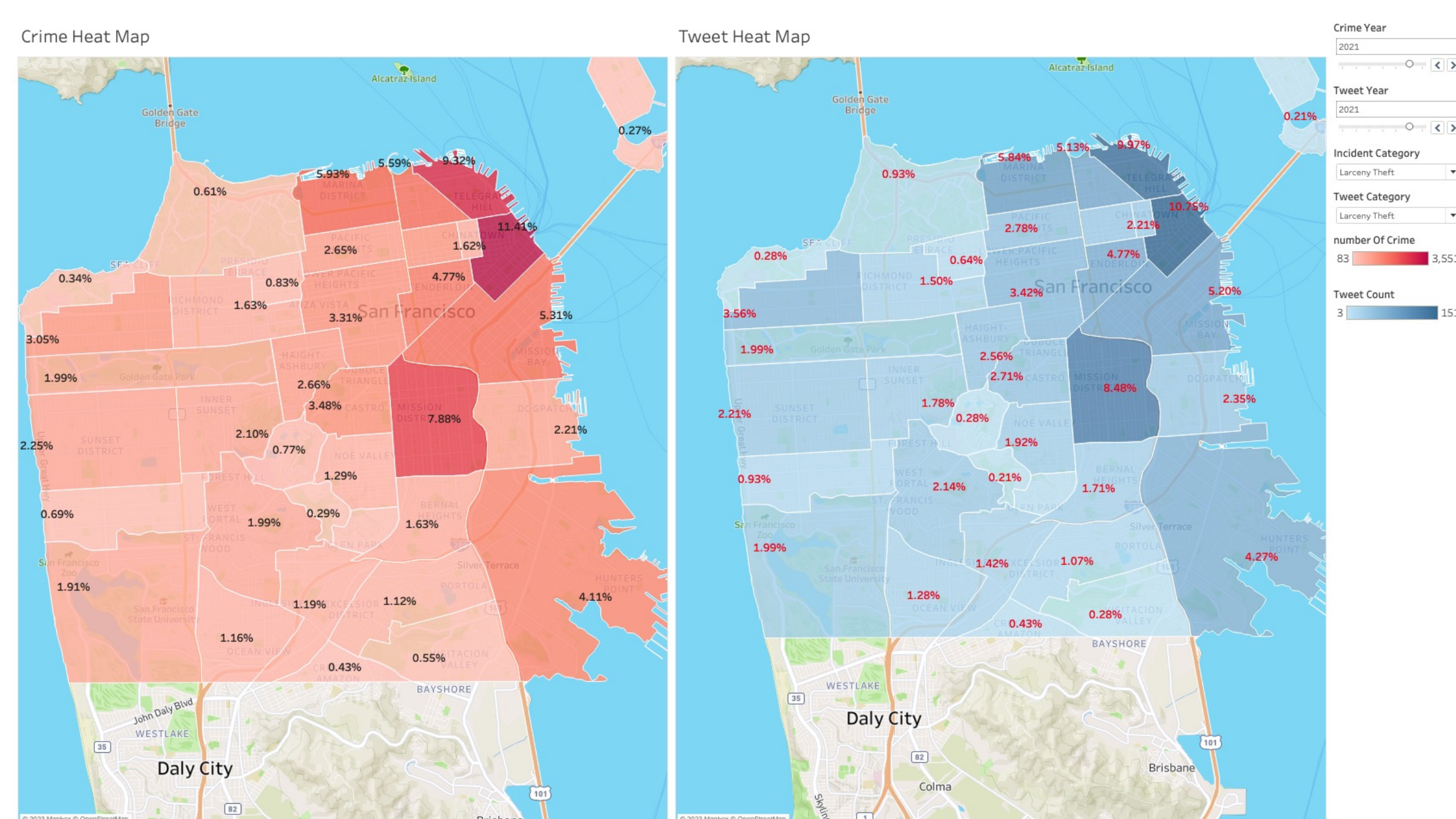
Introduction

- **Problem Domain:** Social Media platforms (e.g., Twitter (renamed as X), Facebook, etc.) are used heavily by public to provide news, opinions, and reactions towards events or topics. Integrating such data with the event or topic factual data could provide a more comprehensive understanding of the underlying event or topic.
- **Our Solution:** We present our visual analytics tool, called **VC-FaT**, that integrates peoples' tweet data regarding crimes in San Francisco city with the city factual crime data. VC-FaT provides a number of interactive visualizations using both data sources for better understanding and exploration of crime activities happened in the city during a period of five years (January 01, 2018 to December 12, 2022).

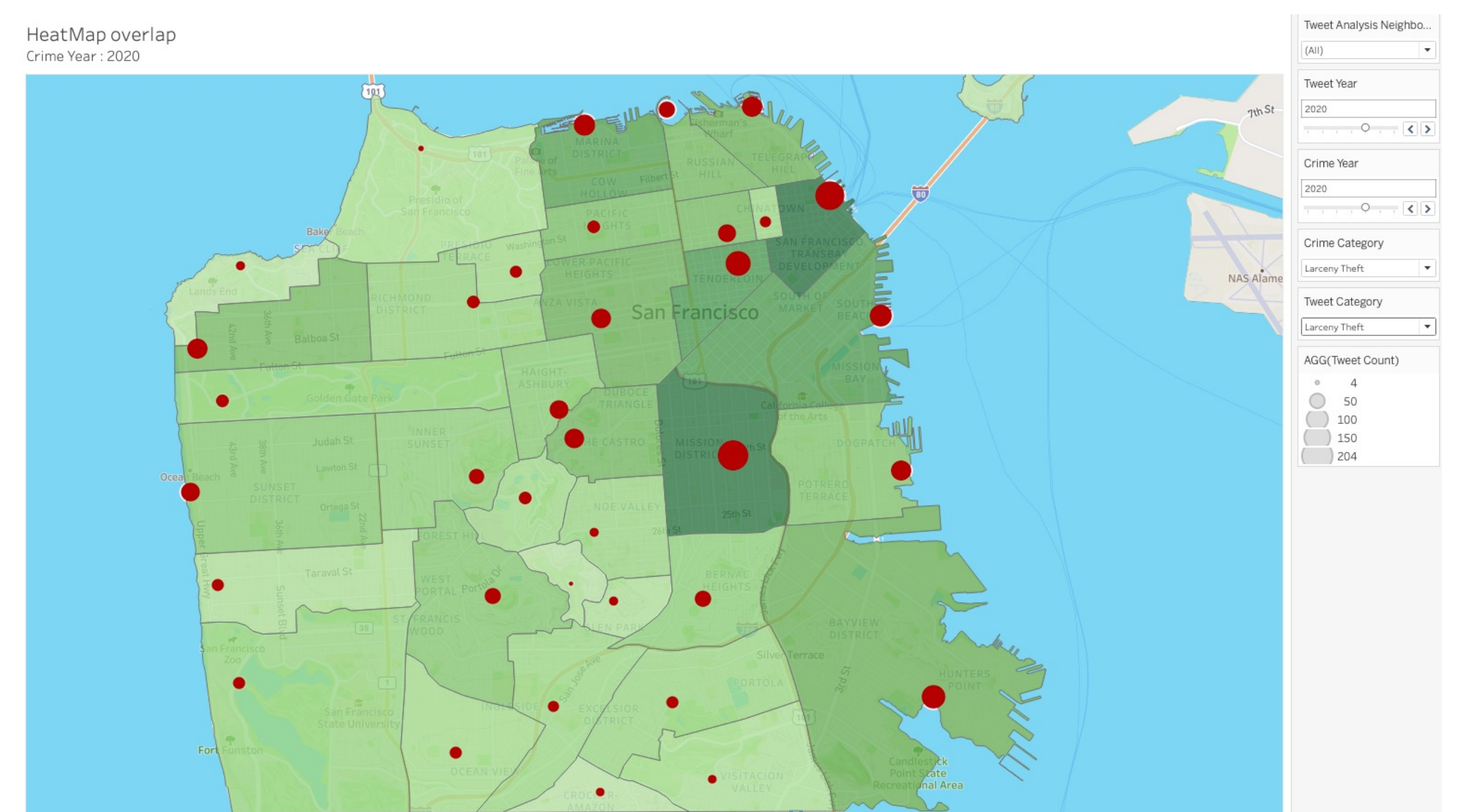
The Datasets

- **Tweet Data:**
 - We used Twitter API1 and collected the tweets based on crime related keywords between January 1, 2018, and December 12, 2022.
 - The tweets were collected, **45,353** tweets, based on either having the San Francisco's districts geolocations or having the crime related keywords with mentioning San Francisco or one of its district in the tweet.
- **San Francisco Factual Crime Data:**
 - The SF crime data was collected from the official website of the SF Government, provided by the SF Police Department (SFPD), and covers the same period of January 1, 2018, to December 12, 2022
 - The initial dataset comprised of 602,901 classified into various categories (e.g., arson, theft, burglary, assault, fraud, robbery, car theft, etc.). After removing the minor traffic violence records (e.g., signal crossing tickets, roadside tickets, etc.), we came up with **401,449** recorded crime incidents.
 - The dataset contains 34 columns of information, including the neighborhood area, incident category, latitude, longitude, police district, and the time the incident was reported and occurred.
- In preprocessing the tweet data, we used **Google Maps API** to map tweets to their corresponding districts in San Francisco. We used the **NLTK Toolkit** to tokenize the keywords so to find out the crime type associated to each tweet

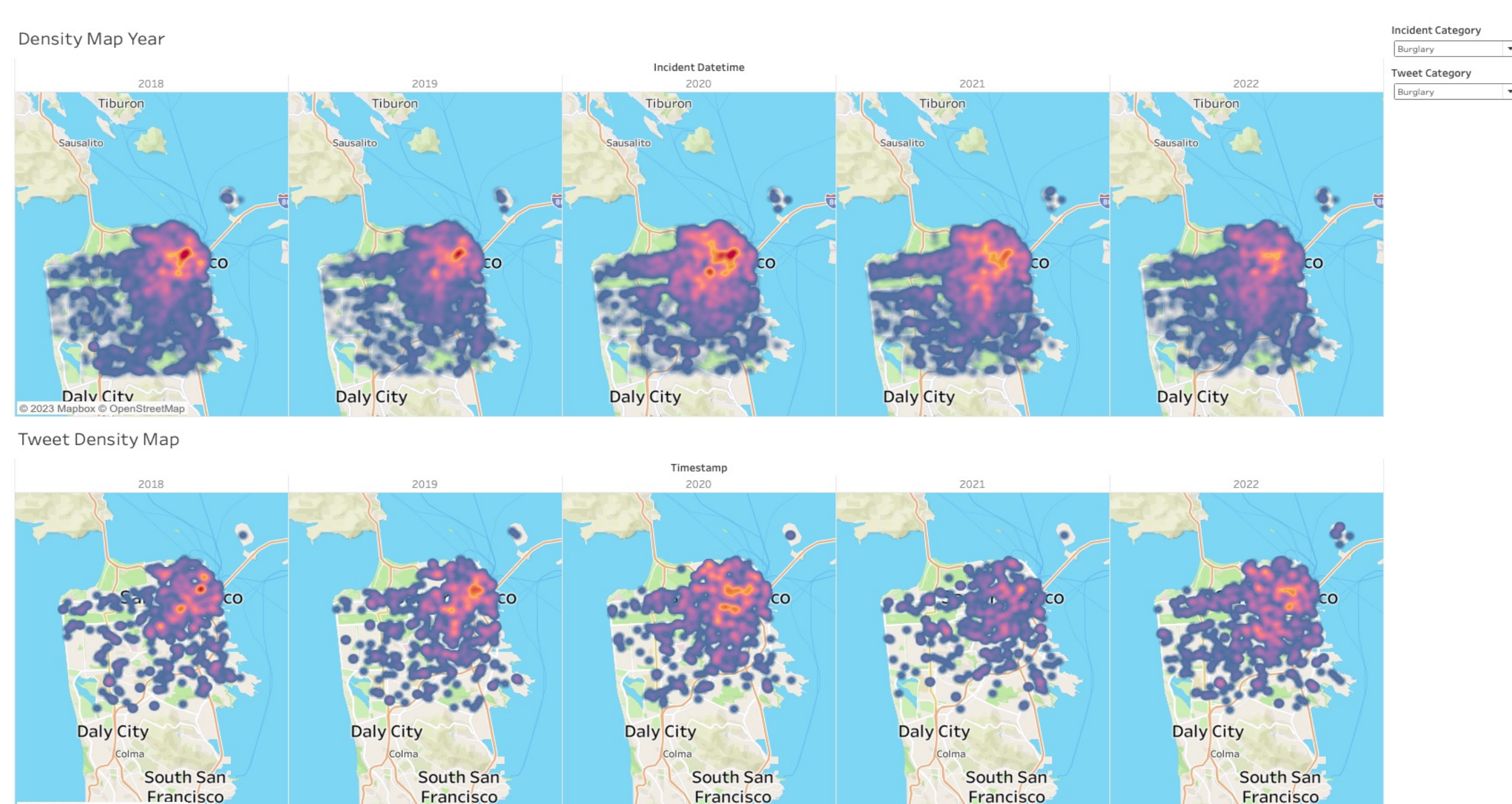
VC-Fat Resulting Visualizations



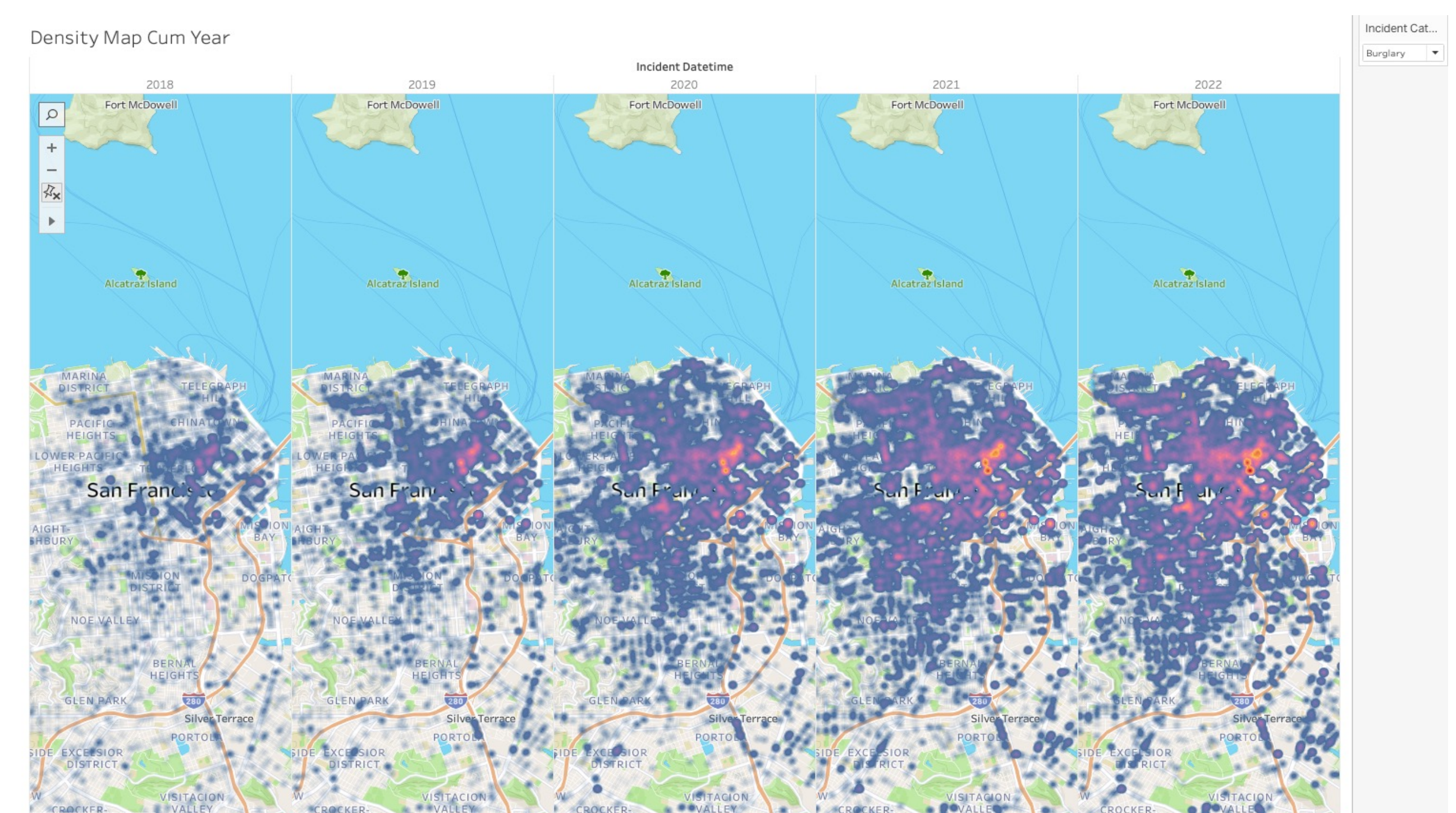
Left Heatmap shows San Francisco factual crime rate while right-side heatmap shows the tweet data, both between the period of Jan. 01, 2018 to Dec. 12, 2022.



Heatmap showing the SF crime data while overlaid bubbles show the associated tweet data



Top density charts show year-wise timeline of SF factual crime data while lower density chart show year-wise tweet data to each SF district.



Accumulative evolution of crime records in SF adding the crime counts from previous years to each next year.